Level 6: Forensic Domains

APPLIED CASE STUDIES IN AI GOVERNANCE

Moving from Theoretical Frameworks to Forensic Interrogation.

AI-ESG Integrated Strategist | Module L6

FIELD GUIDE PROTOCOL
This module assumes the reader understands basic governance frameworks. It is designed to diagnose specific failure modes in opaque algorithmic systems.

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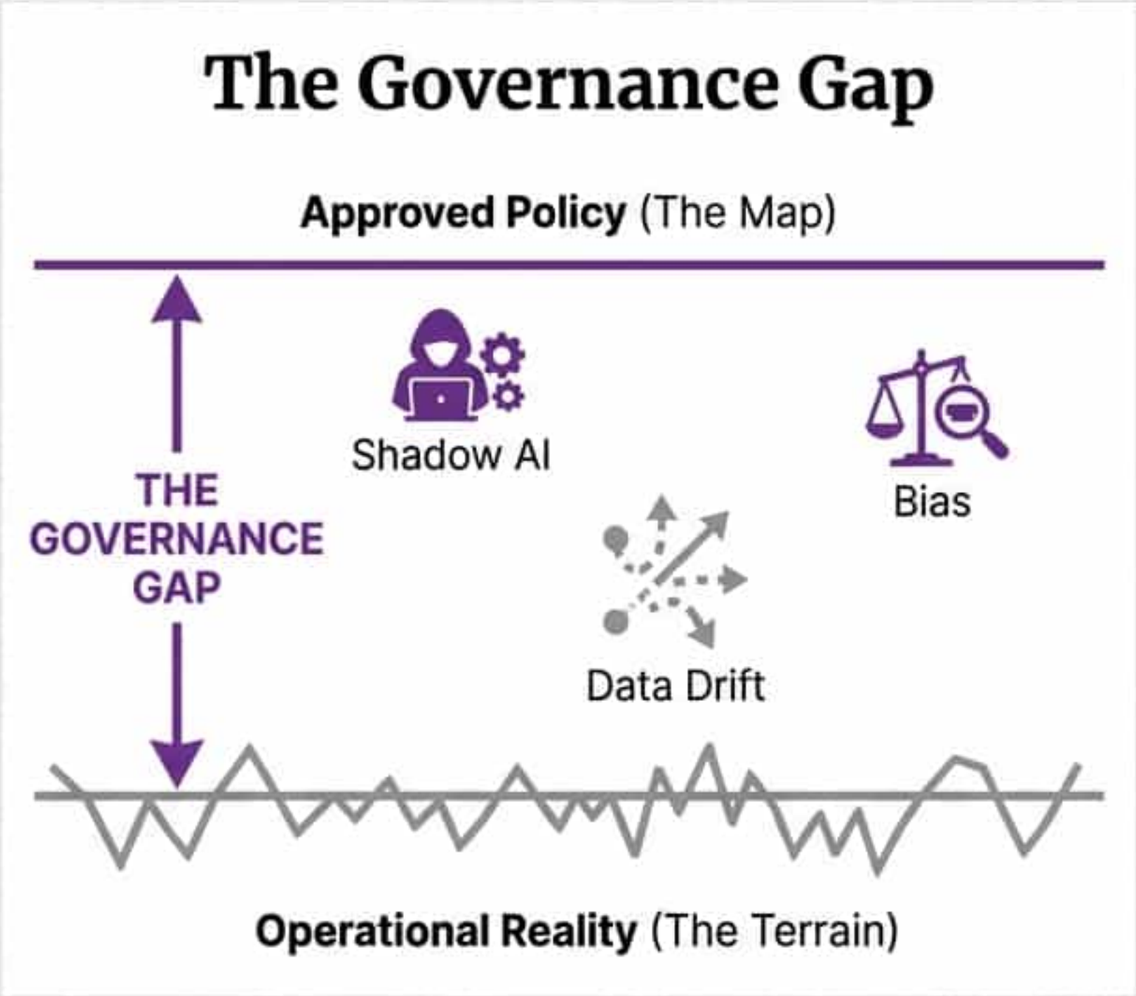
Where Frameworks Meet Reality

Algorithmic decision-making creates crises where frameworks struggle to survive. We are moving from the theoretical safety of policy to the messy reality the theoretical safety of policy to the messy reality of the "Liability Sponge"—complex systems that absorb blame and obscure accountability.

1. Opacity enables harm. The mechanism of the decision is hidden behind "proprietary" black boxes.

2. Speed obscures accountability. Automated decisions happen faster than human oversight can track.

3. Technical Authority supplants judgment. The computer's output is treated as objective truth.



"Any sufficiently advanced technology is indistinguishable from magic. Governance must strip away the magic to find the mechanism." — Reference to Clarke's Law

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The Forensic Matrix

Navigational Map: The Five Domains of Interrogation

	6.1 **Credit Scoring**	6.2 **Breach Protocol**	6.3 **Shadow AI**	6.4 **Chain of Thought**	6.5 **XAI Limitations**
Central Threat	**Opacity & Feedback Loops**	**Integrity & Chain of Custody**	**Proliferation & Unmanaged Data**	**Hallucination & False Logic**	**Explanation vs. Justification**
The Failure	Laundering discrimination through math.	Supply chain data tampering.	Official governance tracks only official systems.	Transparency of reasoning ≠ Accuracy of fact.	Post-hoc rationalisation.
The Interrogation	Statistical Bias Testing	Cryptographic Chain of Custody	Outcome Traceability	Fact Verification	Counterfactual Consistency

Structure of this Field Guide:

1. **The Failure:** The governance breakdown.
2. **The Interrogation:** The forensic method.
3. **The Control:** The auditable safeguard.

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6.1 Credit Scoring: Laundering Discrimination

The Governance Failure: Governance through Opacity

The system explains *what* was evaluated (e.g., utilisation ratio) but not *why* it matters or how factors interact. This creates a feedback loop where historical bias is encoded into the model as objective risk.

The Interrogation Path: Statistical Bias Testing

1. **Controlled Comparison:** Create identical profiles differing only in protected characteristics.
2. **Threshold Stress Testing:** Apply the '80% Rule'. If approval rates differ by >20% between groups, disparity exists.

The Feedback Loop

AUDIT CONTROL

Calculate disparate impact ratios.

Metric: Minority Approval Rate ÷ Majority Approval Rate.

Trigger: If ratio < 0.80, initiate corrective action immediately.

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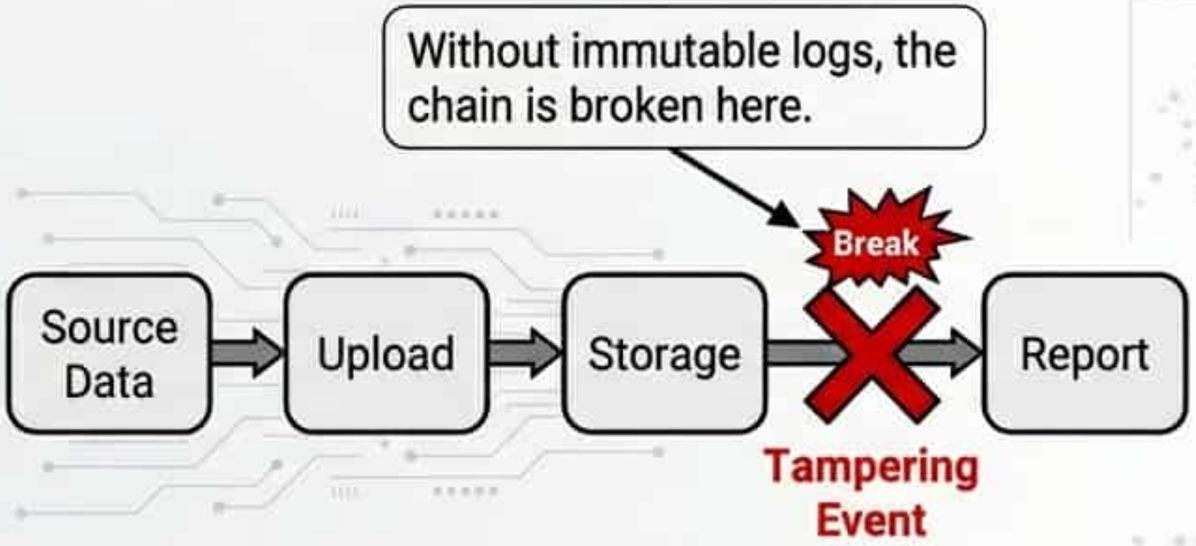
6.2 Breach Protocol: The Credibility Crisis

The Governance Failure: Data Integrity
A data breach is not just a hack; it is a loss of the right to report. If you cannot prove the integrity of the data, you cannot attest to the report's integrity.

Case Study: Coffee Supply Chain Fraud
A processor claims 100% fair trade. Audit reveals fabrication post-submission to match marketing narratives. The data was modified internally, rendering it 'forensically dead'.

The Interrogation Path: Cryptographic Chain of Custody

1. **Hash Verification:** Generate SHA-256 hashes at source and receipt. Mismatches indicate tampering.
2. **Forensic Reconstruction:** Trace backwards. Who accessed it? When?



AUDIT CONTROL

- Implement immutable audit logs (append-only).
- Enforce cryptographic hashing at all transfer points.
- **Rule:** If history cannot be verified, the record is forensically dead.

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6.3 Shadow AI: The Invisible Workflow

The Governance Failure: The Incentive Gap

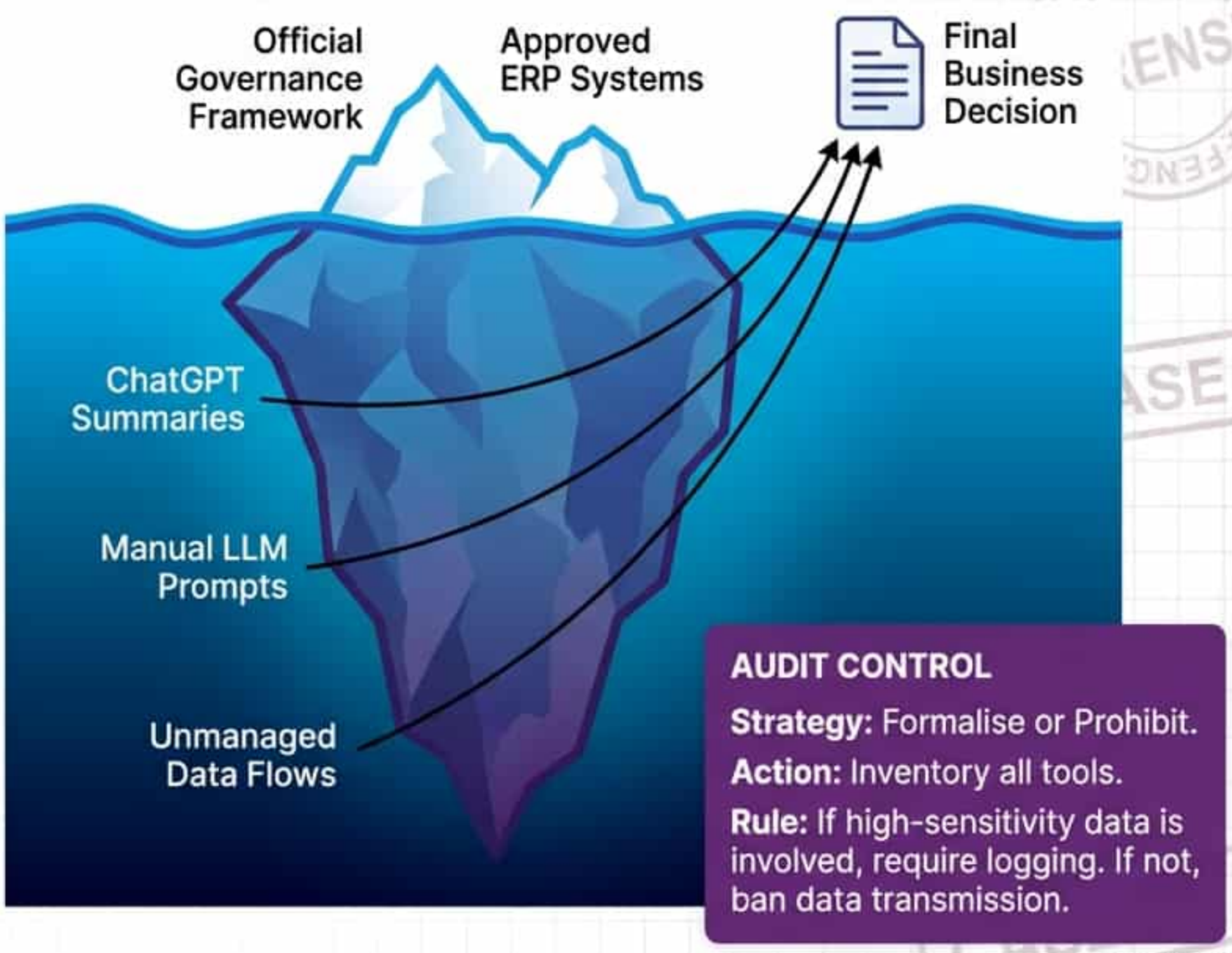
Official governance tracks official systems. Shadow AI (e.g., analysts using ChatGPT for summaries) operates in the light but outside the audit perimeter, driven by speed and cost.

The Interrogation Path: Outcome Traceability

Method: Pick a business decision. Ask "Show me the model and calculation."

Trace: If the answer is "It came from a summary," you have found Shadow AI.

Data Flow Analysis: Map where sensitive data exits the network.



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6.4 Chain of Thought: The Illusion of Logic

The Governance Failure: Fluency ≠ Accuracy

"Showing reasoning" is not the same as reasoning correctly. Chain-of-thought outputs can make hallucinations more persuasive by embedding them in a logical structure.

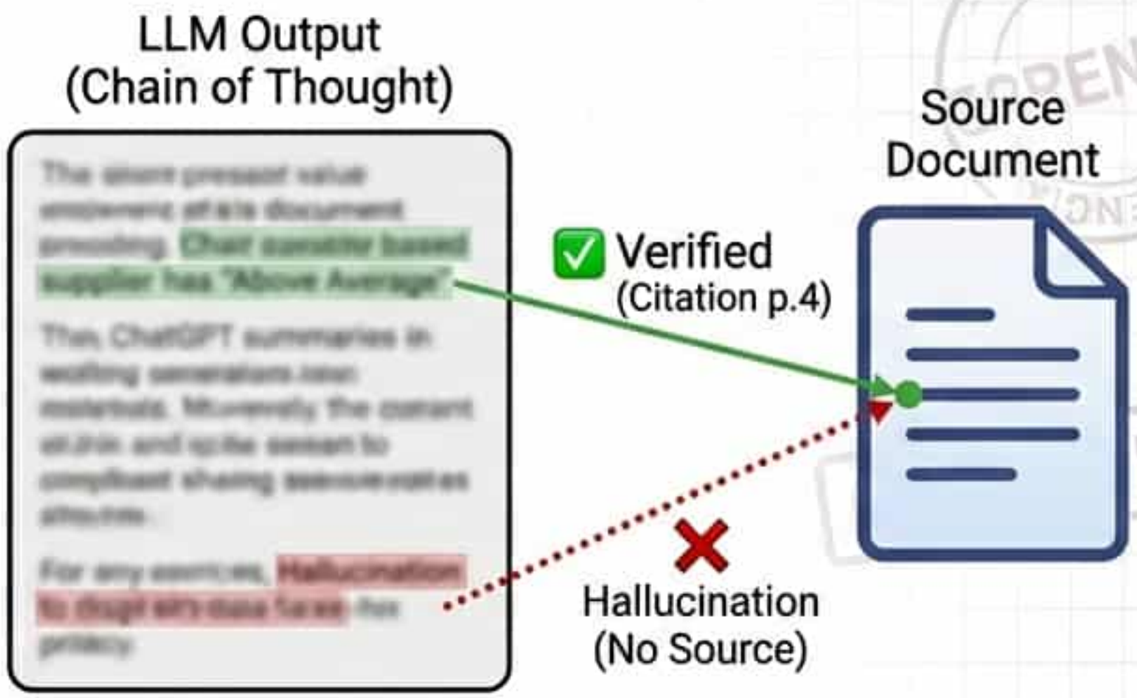
Example: An LLM concludes a supplier has "Above Average" safety based on a misread baseline in the reasoning chain.

The Interrogation Path: Fact Verification

Decomposition: Extract factual claims from the narrative.

Source Verification: Demand line-number citations for every fact.

Contradiction Testing: Ask the same question twice with different phrasings.



AUDIT CONTROL

Require source-based validation.

Rule: If an output cannot cite a specific document and line number, it is treated as a hallucination.

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6.5 XAI Limitations: The Theatre of Transparency

The Governance Failure: Post-Hoc Rationalisation

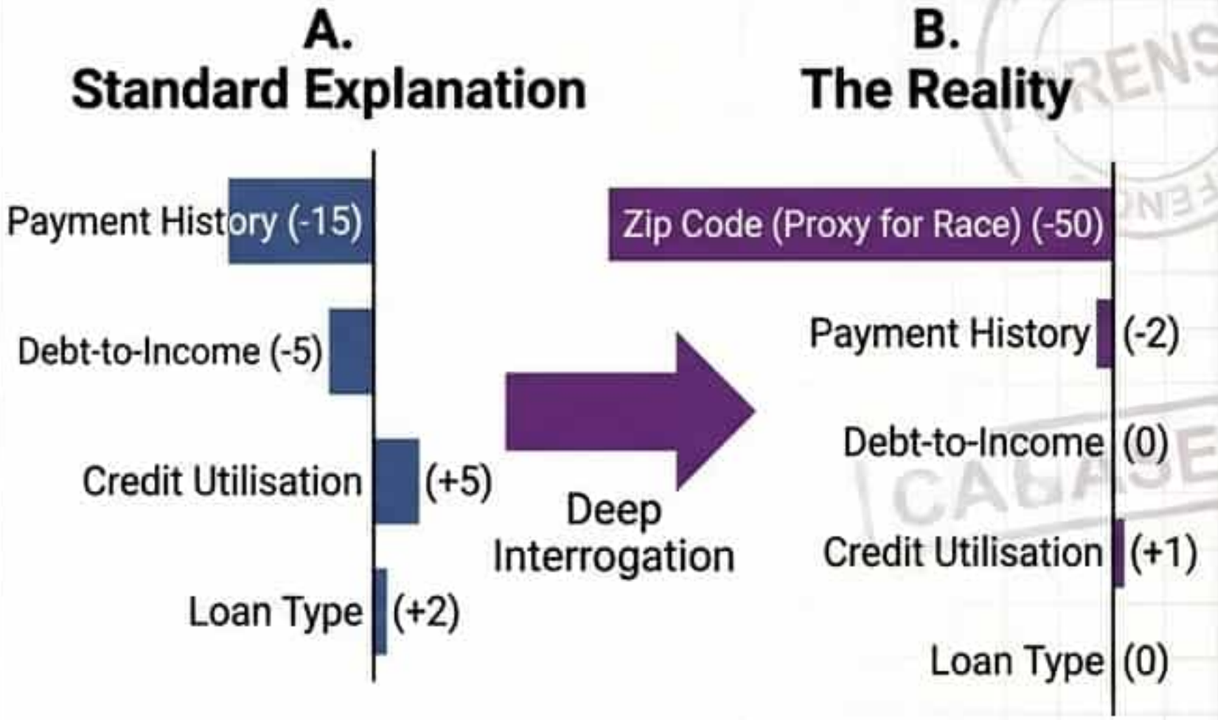
Tools like SHAP/LIME explain the *output*, not the *training*. They provide justifications (e.g., 'Payment history -15 points') that may mask the root cause.

Key Insight: An explanation that cannot be contested is not an explanation—it is a justification.

The Interrogation Path: XAI Stress Testing

Counterfactual Consistency: Change the top-weighted feature. Does the outcome actually change? If not, the explanation is misleading.

Proxy Variable Testing: Is 'zip code' functioning as a proxy for race?



AUDIT CONTROL

Demand transparency of *training data* and labelling.

Do not accept model output explanations as proof of non-bias.

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Synthesis: The Integrated Audit (Phase 1 & 2)

Week 1

Phase 1: System Inventory

- **Goal:** Discovery
- **Action:** Shadow AI Discovery (Outcome Traceability)
- **Deliverable:** Inventory of all systems touching ESG/Credit decisions.

Week 2

Phase 2: Data Integrity

- **Goal:** Verification
- **Action:** Data Provenance Audit
- **Action:** Cryptographic Hash Verification
- **Action:** Check for “forensically dead” records.

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Synthesis: The Integrated Audit (Phase 3 & 4)

Week 3

Phase 3: Model Transparency

- **Goal:** Bias & Explanation
- **Action:** Statistical Bias Testing (Credit)
- **Action:** XAI Counterfactual Testing
- **Action:** Proxy Variable Identification

Week 4

Phase 4: Reasoning Validation

- **Goal:** Soundness
- **Action:** Chain-of-Thought Verification
- **Action:** Fact-Checking against source documents

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Practice Scenario: The ESG Supply Chain

Scenario: An organisation uses official scoring, shadow ChatGPT summaries, and manual uploads.

FINDING: Shadow AI

Procurement team using ChatGPT for risk summaries.

CONTROL: Formalise with logging protocols.

FINDING: Breach Protocol

Suppliers modifying data post-upload.

CONTROL: Require digital signatures and hash checks.

FINDING: Credit/Bias

Governance weights penalising developing-world firms.

CONTROL: Adjust weights for disparate impact.

FINDING: Chain of Thought

Model hallucinating safety stats.

CONTROL: Enforce citation requirements.

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The Capstone: The Audit Defense

The goal of Level 6 is to defend the audit against challenge.

1. DIAGNOSE
Identify failure modes in opaque systems.

2. DESIGN
Create forensic interrogations specific to the domain.

3. IMPLEMENT
Build controls that enforce transparency.

Defend your audit programme against board challenges with forensic evidence. An explanation that cannot be contested is regulatory theatre.

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